

Brand and Region Graphs

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

brand_io_rows=43

brand_econ_rows=1808

region_rows=147

Interpretation option: IO concentration, promo intensity, and regional dispersion.

Table: Brand_IO_Metrics

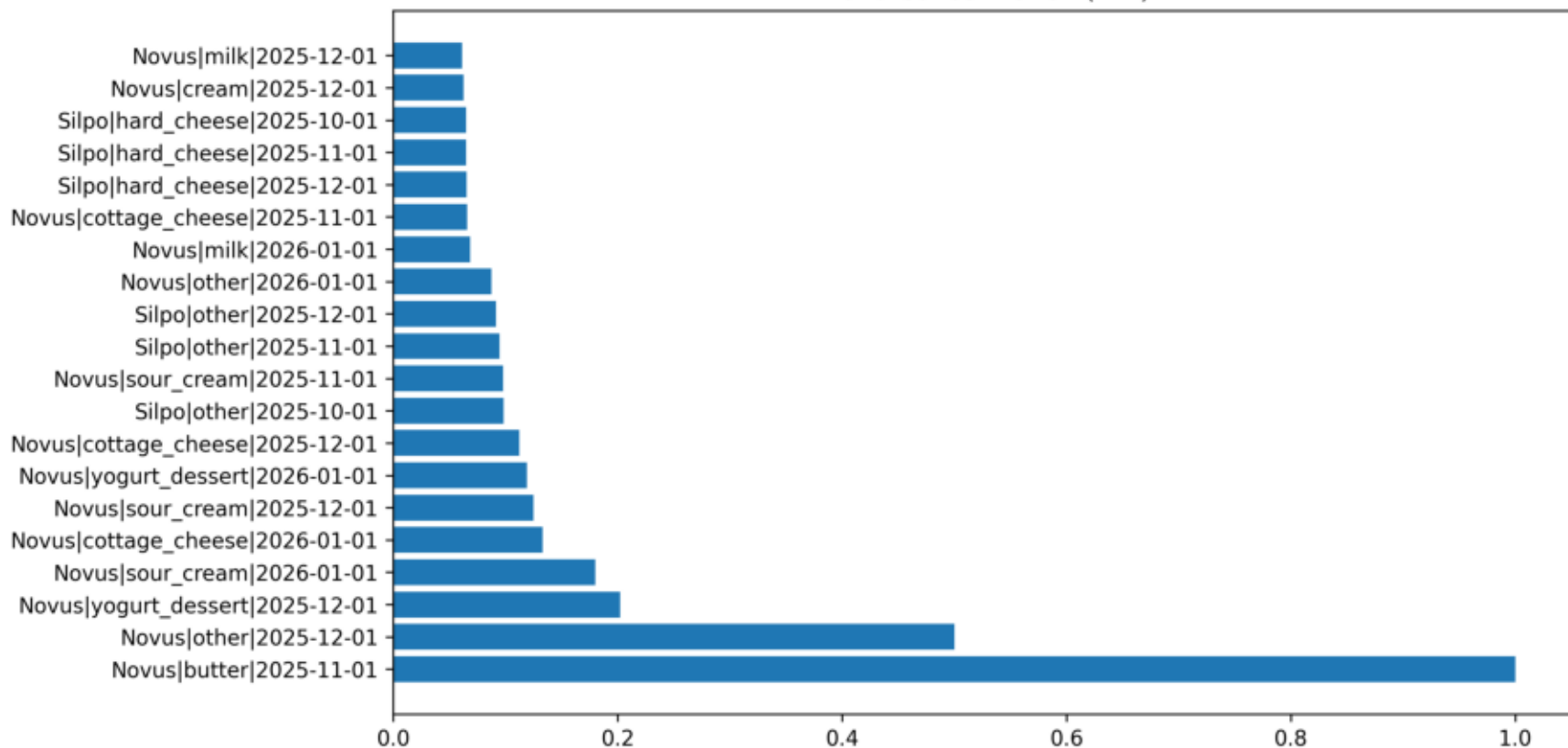
source	standardized_type	month	hhi_brand	top3_share	private_label_share	sku_count	n_obs
Novus	butter	2025-11-01	1.0	1.0	0.0	1	1
Novus	cottage_cheese	2025-11-01	0.06574394463667821	0.3529411764705882	0.014705882352941176	68	68
Novus	cottage_cheese	2025-12-01	0.11224489795918367	0.42857142857142855	0.0	14	14
Novus	cottage_cheese	2026-01-01	0.13314967860422403	0.5454545454545454	0.030303030303030304	33	33
Novus	cream	2025-11-01	0.023989898989898985	0.17424242424242423	0.007575757575757576	132	132
Novus	cream	2025-12-01	0.0625	0.29166666666666663	0.0	24	24
Novus	cream	2026-01-01	0.05115436849253055	0.2553191489361702	0.0	47	47
Novus	hard_cheese	2025-11-01	0.01761395164486722	0.13793103448275862	0.0022988505747126436	435	435
Novus	hard_cheese	2025-12-01	0.04064272211720225	0.2173913043478261	0.0	46	46
Novus	hard_cheese	2026-01-01	0.04146652288991276	0.25742574257425743	0.009900990099009901	100	101
Novus	milk	2025-11-01	0.03230653643876786	0.19008264462809918	0.0	120	121
Novus	milk	2025-12-01	0.06122448979591835	0.2857142857142857	0.0	21	21
Novus	milk	2026-01-01	0.06823979591836733	0.3392857142857143	0.0	56	56
Novus	other	2025-11-01	0.060815326353308094	0.3283582089552239	0.0	67	67
Novus	other	2025-12-01	0.5	1.0	0.0	2	2
Novus	other	2026-01-01	0.08723599632690543	0.393939393939394	0.0	33	33
Novus	sour_cream	2025-11-01	0.09765625	0.4375	0.0	32	32
Novus	sour_cream	2025-12-01	0.125	0.375	0.0	8	8
Novus	sour_cream	2026-01-01	0.18000000000000002	0.65	0.0	20	20
Novus	yogurt_dessert	2025-11-01	0.0484078684881695	0.2814070351758794	0.01507537688442211	199	199
Novus	yogurt_dessert	2025-12-01	0.20226843100189035	0.7391304347826086	0.0	23	23
Novus	yogurt_dessert	2026-01-01	0.11905839746491623	0.5106382978723404	0.0	47	47
Silpo	cottage_cheese	2025-10-01	0.04730395378755094	0.26366559485530544	0.025264124942581533	208	2177
Silpo	cottage_cheese	2025-11-01	0.04717982132060342	0.25992097577735784	0.02576876825287751	211	5821
Silpo	cottage_cheese	2025-12-01	0.04837308571635941	0.26581325301204817	0.030873493975903613	196	1328
Silpo	cream	2025-10-01	0.02934070920989699	0.1911013858497447	0.016046681254558718	259	2742
Silpo	cream	2025-11-01	0.028471594126903442	0.1863520579166108	0.016087947446038343	267	7459
Silpo	cream	2025-12-01	0.029120948162431992	0.19133996489174954	0.016383850204798128	248	1709
Silpo	hard_cheese	2025-10-01	0.06499127867281239	0.362760834670947	0.052969502407704656	116	1246
Silpo	hard_cheese	2025-11-01	0.06502512501091798	0.36244158878504673	0.052570093457943924	118	3424
Silpo	hard_cheese	2025-12-01	0.0653136485469618	0.37157107231920206	0.05236907730673317	118	802
Silpo	milk	2025-10-01	0.028914814375894944	0.1885317115551694	0.014335360556038228	434	4604
Silpo	milk	2025-11-01	0.02981149983943769	0.1924998017918021	0.014191706969000238	444	12613
Silpo	milk	2025-12-01	0.03008707808449764	0.1943874058863792	0.014373716632443531	425	2922
Silpo	other	2025-10-01	0.09804676817006609	0.4928813559322034	0.022372881355932205	135	1475
Silpo	other	2025-11-01	0.0946189970877688	0.4821294676330917	0.022494376405898527	142	4001
Silpo	other	2025-12-01	0.09126479484519621	0.4715969989281886	0.022508038585209004	135	933
Silpo	sour_cream	2025-10-01	0.03526736960042432	0.2064834390415786	0.015503875968992248	137	1419
Silpo	sour_cream	2025-11-01	0.03590539901169873	0.20901093041855506	0.01626233004532125	134	3751
Silpo	sour_cream	2025-12-01	0.03626850212168799	0.21040462427745665	0.016184971098265895	125	865

Table: Prozorro_ByRegion

region	product	standardized_type	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
-	Інше/невідомо	other	123	0	nan	5.262032520325205	5.08	0.7708234465939691	3.05	7.38	4.355000000000001	4.720000000000001	5.94	6.538	0.22578340144087045	0.04304271352944822	0.146487777036074	0.11183743324997715	0.10551139923428733
-	Вершки	cream	31	0	nan	317.1074193548387	289.0	87.12171256228238	198.0	598.8000000000002	229.5	259.0	338.97	498.0	1.6916176195835828	3.3619193930656515	0.2747388022000028	0.3118541073650984	0.3571232254712644
-	Молоко питне	milk	34	0	nan	62.38529411764706	43.08	33.71094078631968	23.34	120.0	24.531	36.69	97.19999999999999	118.05	0.48040246084400523	-1.5375788259205696	0.5403667845621938	0.8050641530545013	0.7786971954286577
-	Сир кисломолочний	cottage_cheese	11	0	nan	118.50636363636363	126.96	20.29961884998202	85.0	150.0	89.9	104.1	129.5	145.825	-0.1764338812511516	-0.8821627282689182	0.17129560157858975	0.26051494187992635	nan
-	Сир твердий	hard_cheese	46	0	nan	269.9910869565217	258.18	49.14078085701035	214.0	475.0	225.0	248.95499999999998	270.75	385.0	2.671068923901233	8.054536749966092	0.18200890040834491	0.1964883783704427	0.2298474616605193
-	Сметана	sour_cream	30	0	nan	105.84899999999999	100.08	22.27987438056582	57.0	149.0	79.84800000000001	95.685	112.5	146.1	0.5283846865444983	0.2164345365729874	0.2104873393283434	0.32759822957750356	nan
Івано-Франківська	Інше/невідомо	other	77	0	nan	6.13909090909091	6.22	0.8565355276039908	4.4	7.7	4.88	5.39	6.76	7.420000000000001	-0.06332253777988718	-1.0671678869465335	0.13952155788011103	0.13501142195235818	0.14075689337226976
Івано-Франківська	Вершки	cream	70	0	nan	375.10585714285713	358.8	66.26840947376205	279.0	580.2	298.35	320.0	418.5	484.60499999999996	1.0069550722882605	0.9503953763013788	0.17666588834021876	0.23052262164322257	0.2094621469060926
Івано-Франківська	Молоко питне	milk	91	0	nan	41.87626373626373	38.64	18.929264330856814	27.24	144.0	29.04	33.06	45.0	52.95	4.314563207421908	20.527462098358995	0.45202849160739655	0.3312028275900505	0.3688211584291134
Івано-Франківська	Сир кисломолочний	cottage_cheese	29	0	nan	123.81655172413794	117.0	20.315498003083764	86.04	154.8	94.6	111.0	145.0	151.992	0.028972545093300342	-1.1887327816900202	0.1640774009628898	0.2394727137499849	nan
Івано-Франківська	Сир твердий	hard_cheese	19	0	nan	299.4673684210526	291.0	40.40455376454838	239.0	366.0	239.576	276.0	325.5	361.30199999999996	0.30297193722009097	-0.9177907552914077	0.1349213905260602	0.1365686406707588	nan
Івано-Франківська	Сметана	sour_cream	17	0	nan	119.53176470588235	123.0	17.123022599447108	92.94	154.4	96.588	108.0	135.0	141.28	0.2375463977830526	-0.752653869866275	0.14325081405415288	0.17484610644693843	nan
Волинська	Інше/невідомо	other	122	0	nan	6.022704918032787	6.0	0.8384056463964787	4.350000000000001	8.120000000000001	4.702500000000001	5.41	6.7	7.294999999999999	-0.0006848267387692877	-0.7086693790373868	0.13920749195036597	0.1154776134492342	0.11044806372942716
Волинська	Вершки	cream	95	0	nan	393.6588421052632	388.0	67.05673117184827	220.0	575.0	270.0	349.65	447.18	489.09	0.027175912612710983	0.10689690456314738	0.1703422456186504	0.224615991305283	0.217872825283622
Волинська	Молоко питне	milk	91	0	nan	41.653186813186814	40.0	15.268738424136854	26.58	170.58	29.22	36.06	44.1	53.5	6.835106032985606	57.4192874675157	0.36656831307089777	0.30598775953930835	0.3478385318273773
Волинська	Сир кисломолочний	cottage_cheese	7	0	nan	103.64285714285714	95.0	31.49570643452094	84.0	174.0	84.87	88.45	97.8	151.67999999999995	2.4817734189143517	6.34943272263103	0.3038868987479622	nan	nan
Волинська	Сир твердий	hard_cheese	9	0	nan	278.2922222222222	296.22	29.208462719637343	215.05	300.0	230.23000000000002	265.2	299.0	299.976	-1.487151560011889	1.783268234978916	0.10495608711735309	0.16174524869261295	nan
Волинська	Сметана	sour_cream	16	0	nan	91.80625	82.9	23.64615454994744	70.0	159.0	71.5	75.0	102.3	130.875	1.7372893901484199	3.32382893399822	0.2575658470958942	0.3050119100803857	nan
Вінницька	Інше/невідомо	other	224	0	nan	5.9834375	5.92	0.9434738147535205	4.4	9.5	4.5915	5.2	6.705	7.5	0.34045959941789666	-0.27200527089185256	0.15768090077877817	0.1167961230941666	0.11985987941238893
Вінницька	Вершки	cream	155	0	nan	388.7090967741936	399.96	89.42348235548359	220.0	600.0	232.1	331.5	436.95000000000005	513.09	-0.306501915658805	-0.716643477879531	0.23005214569306388	0.3349021388284887	0.34161777093102536
Вінницька	Молоко питне	milk	219	0	nan	52.2765296803653	45.0	25.28106376617377	26.94	156.0	33.762	40.41	48.0	123.0	2.4834776111116184	5.220204583871496	0.48360256353568093	0.41877458029545966	0.42705496005063925
Вінницька	Сир кисломолочний	cottage_cheese	41	0	nan	154.3029268292683	136.68	45.40121882966924	95.0	240.0	97.9	115.08	209.34	209.34	0.26665026209047643	-1.5337564043630258	0.29423433348029987	0.43088366161487685	0.42809494842050094
Вінницька	Сир твердий	hard_cheese	64	0	nan	304.541875	296.85	49.18356680892125	226.2	400.0	240.0	272.4	338.25	396.72	0.44176423899093265	0.7465605528262551	0.1615001773037657	0.23608461491535268	0.23499022676704917
Вінницька	Сметана	sour_cream	24	0	nan	130.06916666666666	132.42000000000002	18.2396777042146	86.2	168.0	105.045	119.25	138.81	163.54199999999997	-0.0022237530371951976	0.9158513313054573	0.14023060323710795	0.17171791622073312	nan
Дніпропетровська	Інше/невідомо	other	171	0	nan	5.713684210526315	5.7	0.927482928168138	3.95	8.4	4.234999999999999	4.9350000000000005	6.35	7.2	0.263344053032623	-0.5094998899364347	0.16232659943989153	0.18844437419561066	0.18109162198920836
Дніпропетровська	Вершки	cream	217	0	nan	364.1723041474654	388.5	124.23589206454683	105.0	710.0	165.6	270.0	440.0	560.1	0.08194635054649219	-0.34251438345771357	0.3411459099158722	0.5169727747499001	0.5196851945398626
Дніпропетровська	Молоко питне	milk	214	0	nan	45.25275700934579	42.0	16.68886385798194	27.9	168.65	30.878999999999998	35.4	49.8	64.5595	3.805365969874612	21.13943826216013	0.368792200982037	0.39138638000350007	0.40342438658858254
Дніпропетровська	Сир кисломолочний	cottage_cheese	53	0	nan	124.83433962264151	110.0	60.898508302731216	60.0	260.0	60.0	69.0	170.4	228.66	0.4752997250366984	-1.0791851007442443	0.4878347585231378254	0.6835941796123562	0.6317646106187046
Дніпропетровська	Сир твердий	hard_cheese	56	0	nan	268.1533928571429	240.0	99.03465873166526	117.0	500.0	148.0	188.0	325.5	448.0	0.6285523716725998	-0.5475637308874752	0.3693209236566527	0.5663857212987943	0.6374315511203323
Дніпропетровська	Сметана	sour_cream	58	0	nan	120.485	127.02000000000001	39.29082028632044	65.52	210.0	71.75399999999999	77.7	149.25	180.0	0.13051527773069538	-1.2389449846515785	0.32610549268639616	0.5212954271666449	0.5143143159682037
Донецька	Інше/невідомо	other	6	0	nan	8.1	7.699999999999999	3.4578895297565535	4.9	13.05	4.950000000000001	5.150000000000001	10.1375	12.325000000000001	0.4076514311186301	-1.9347565898317454	0.42689994194525355	nan	nan
Донецька	Вершки	cream	10	0	nan	395.34	418.0	66.80131735227981	229.0	460.8	276.745	400.625	423.75	449.19	-2.0514243617057843	4.445271286055433	0.1689718150257495	0.20641447381018854	nan
Донецька	Молоко питне	milk	6	0	nan	56.175000000000004	47.1	23.25071504276804	36.85	99.0	38.1375	42.0	61.8	90.5	1.5812724091974741	2.3490225486824254	0.4138979090835432	nan	nan
Донецька	Сир кисломолочний	cottage_cheese	1	0	nan	169.84	169.84	0.0	169.84	169.84	169.84	169.84	169.84	169.84	nan	nan	0.0	nan	nan
Донецька	Сир твердий	hard_cheese	4	0	nan	339.5	350.0	66.93529213601248	259.0	399.0	266.65	297.25	392.25	397.65	-0.4621657318097252	-3.2396287376208193	0.19715844517234898	nan	nan
Донецька	Сметана	sour_cream	1	0	nan	93.11	93.11	0.0	93.11	93.11	93.11	93.11	93.11	93.11	nan	nan	0.0	nan	nan
Житомирська	Інше/невідомо	other	267	0	nan	5.431685393258427	5.5	0.8881943785196804	3.93	7.85	4.1930000000000005	4.600000000000001	6.0	6.837	0.25756165945633136	-0.8293231227844498	0.16352095421838475	0.10641595291155065	0.10748954401491716
Житомирська	Вершки	cream	144	0	nan	381.2891666666667	378.0	66.19066741201173	250.0	546.0	272.15	328.75	435.5	470.0	-0.09165049127618276	-0.9948926277975825	0.17359703133101972	0.24260603500875766	0.2428281221341698
Житомирська	Молоко питне	milk	176	0	nan	48.41715909090909	39.6	24.979674111970713	22.14	201.0	35.815	38.985	43.925	97.025	3.554666661139167	14.807257373659402	0.5159260597068148	0.3625857047011054	0.38194157148194957
Житомирська	Сир кисломолочний	cottage_cheese	79	0	nan	182.0755696202532	192.0	24.28537022660284	109.92	233.9	121.08	183.15	192.0	198.12	-1.7783129904173378	2.5373832775235265	0.13338071811201108	0.14876720342872612	0.14493452727899134

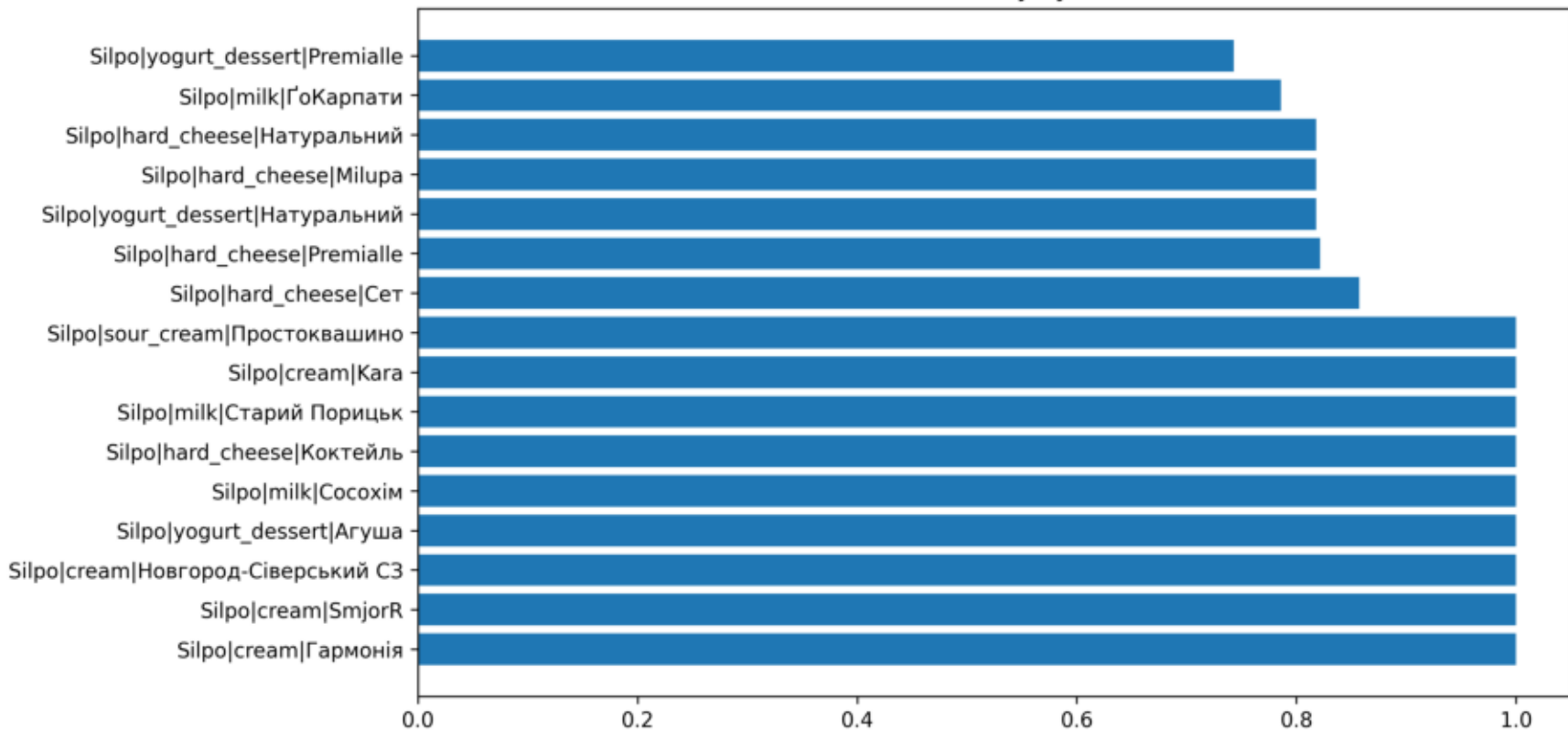
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Brand Concentration (HHI)



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Promo Intensity by Brand



ProZorro Median Unit Price by Region/Product

